



Biodegradable poly(ester)-poly(methyl methacrylate) copolymer for marine anti-biofouling

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ABSTRACT

Being biodegradable and eco-friendly, polyesters are promising in marine anti-biofouling. However, the slow degradation and poor adhesion limit their applications. In the present work, we have prepared poly(ester)-poly(methyl methacrylate) copolymer via radical ring-opening copolymerization of 2-methylene-1,3-dioxepane (MDO) and methyl methacrylate (MMA). The copolymer coating has good adhesion to substrate due to the incorporation of MMA moieties. The main-chain ester bonds from MDO enable the copolymer to enzymatically and hydrolytically degrade, and the degradation rate increases with the ester content. The polymers can serve as a carrier and controlled release matrix for organic antifoulant 4,5-dichloro-2-octylisothiazolone (DCOIT), and the release rate increases with the ester content. Marine field test shows the system of the copolymer and DCOIT has excellent antifouling performance for more than 3 months.

1. Introduction

It is well known that marine biofouling often has detrimental effects on marine industries and other marine activities [1–3]. Antifouling coatings containing various biocides had long been used to prevent and control marine biofouling, particularly the tin-based self-polishing coatings. However, they were banned in 2008 due to their high toxicity to marine ecology [4]. To construct environmentally friendly anti-fouling systems, much effort has been paid to tin-free acrylate based self-polishing coatings [5–7], silicone based fouling release coatings [8–10], self-generated hydrogels [11–13], and slippery liquid infused surfaces [14,15]. Biodegradable polymers are recently developed anti-fouling materials [16–19]. They can form a dynamic surface that continuously renews and thus reduces the colonization of marine organisms [20]. They can also serve as the carrier and controlled release matrix of antifoulants. The degradable main chains enable the self-renewal of coating surface even at static immersion, quite different from traditional SPCs whose erosion is determined by the washing of flowing water. Biodegradable polymers finally decompose into small molecules in marine environment and do not give rise to the so-called microplastic pollution. Thus, the system consisting of biodegradable polymer and organic antifoulant is eco-friendly.

Note that biodegradable aliphatic polyesters such as polycaprolactone (PCL) and polylactic acid (PLA) can not be directly used in

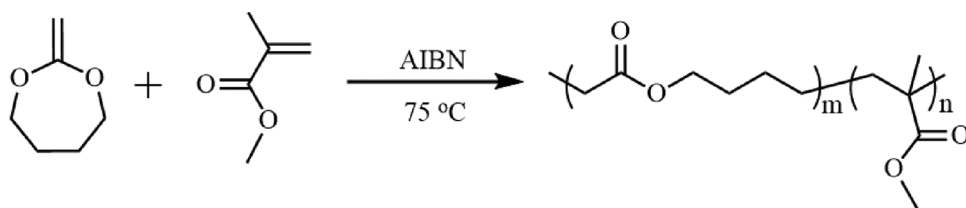
marine antifouling because of their poor adhesion and slow degradation due to high crystallinity [21–23]. Chemical modification [14] or blending [24,25] which lowers their crystallinity can significantly improve their performance in marine anti-biofouling.

On the other hand, polyacrylates are widely applied in coatings and adhesives owing to their remarkable film formation performance and adhesion to various substrates. Yet they are not degradable in marine environment. Biodegradable polyacrylates are promising in marine anti-biofouling because they are expected to exhibit advantages from both polyacrylates and biodegradable polymers. Radical ring-opening polymerization (RROP) of cyclic ketene acetals (CKAs) and vinyl monomers yields poly(vinyl-co-ester)s [26–28]. Such copolymers have low crystallinity due to the branched structure and hence a higher degradation rate [29,30]. Thus, RROP provides a facile and efficient strategy to obtain biodegradable polyacrylates. Compared with other biodegradable polymers such as PCL-based polyurethane [17], biodegradable polyacrylates are easier to synthesize and have better film forming ability. In the present study, we have synthesized poly(ester)-poly(methyl methacrylate) copolymer by RROP of 2-methylene-1,3-dioxepane (MDO) and methyl methacrylate (MMA). We have studied its enzymatic and hydrolytic degradation as well as the release of organic antifoulant. We also have examined its anti-biofouling performance by marine field tests. Our aim is to develop an eco-friendly and effective marine anti-biofouling system.

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Scheme 1. The synthetic route of PMMx.

2. Experimental

2.1. Materials

2-Methylene-1,3-dioxepane (MDO) was synthesized following procedures detailed elsewhere [31]. Methyl methacrylate (MMA) from Aladdin was purified by distillation before use. 2,2-Azobisisobutyronitrile (AIBN) from Aladdin was recrystallized twice from methanol. 4,5-Dichloro-2-octylisothiazolone (DCOIT) was purchased from Baiao Chemical (Dalian, China) and used as received. 1,4-Dioxane was dried by CaH_2 and distilled before use. Artificial seawater (ASW) was prepared following ASTM D1141-98 (2013) [32]. Lipase was purchased from Sigma and used as received. Other reagents were used as received.

2.2. Synthesis of copolymer of MDO and MMA

As shown in Scheme 1, the copolymer was synthesized by RROP in dioxane. Typically, 8.00 g of MMA (80.0 mmol), 2.28 g of MDO (20.0 mmol) and 0.82 g of AIBN (0.5 mmol) were dissolved in 10 mL of dioxane in a Schlenk tube and the reaction mixture was degassed by three freeze-evacuate-thaw cycles. The polymerization was performed under nitrogen at 75 °C for 24 h. The resultant polymer was precipitated in methanol and dried under vacuum at 70 °C overnight. $^1\text{H-NMR}$ (600 MHz, CDCl_3 , ppm): 4.01 ($-\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$), 3.63 ($-\text{COOCH}_3$), 2.48 ($-\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$), 1.2–1.7 ($-\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$). The copolymer is designated as PMMx, where x is the molar percentage of polyester determined by $^1\text{H-NMR}$ (Table 1).

3. Characterization

3.1. Proton nuclear magnetic resonance spectroscopy (^1H NMR)

$^1\text{H-NMR}$ spectra were recorded on a Bruker AV600 NMR spectrometer using CDCl_3 as the solvent and tetramethylsilane as the internal standard.

3.2. Gel permeation chromatography (GPC)

The number-average molecular weight (M_n) and the polydispersity (PDI) were determined by Agilent gel permeation chromatography (1260 Infinity) with refractive index (RI) and ultraviolet (UV) detectors. The experiment was conducted in tetrahydrofuran (THF) at 35 °C using two identical PLgel columns (5 μm , MIXED-C) at a flow rate of 1.0 mL/

min. Calibration was done with a series of narrowly dispersed polystyrene standards (1.30×10^3 to 2.21×10^6 g/mol).

3.3. Differential scanning calorimeter (DSC)

DSC test was performed on a NETZSCH DSC 204F1 differential scanning calorimeter under a nitrogen flow of 50 mL/min. The sample was quickly heated to 150 °C and equilibrated for 5 min to eliminate thermal history, and then it was cooled to -60 °C at a rate of 10 °C/min. Subsequently, it was heated to 150 °C at a rate of 10 °C/min. The glass transition temperature (T_g) was obtained from the endothermic shift during the second heating scan.

3.4. Adhesion test

The adhesion of PMMx coating on a glass fiber reinforced epoxy panel was measured using a pull-off adhesion tester (PosiTest AT-A Automatic). PMMx coating with a dry thickness of ~ 300 μm was prepared via solution casting where the polymer concentration in xylene was 30% (w/v). Pull-test adhesion data were obtained by detaching an aluminum dolly (20 mm in diameter) at a speed of 0.2 MPa/s at 25 °C. Five different points on each sample were tested to obtain an average value.

3.5. Hydrolytic degradation

The PMMx coating on an epoxy resin panel ($20 \times 20 \text{ mm}^2$) was prepared via solution-casting. Generally, solution of polymer in xylene (30%, w/v) was dripped onto the panel and dried in a fume hood at room temperature for 3 days and then under vacuum at 40 °C for 2 days to remove the solvent. The test was conducted in ASW at 25 °C. The initial weight (W_0) of each dried coating together with its panel was measured before the coating was immersed in ASW which was replaced every 2 weeks. At certain intervals, the sample was taken out and rinsed with deionized water. The sample was dried at 25 °C for 1 day and then freeze-dried for 3 days before their weights (W_t) were recorded. The mass loss was calculated by $(W_0 - W_t)/\text{test area}$. For each polymer, three coated panels were prepared and measured to obtain an average value.

The molar mass of degradation product of PMMx was also studied under an accelerated condition. The solution of 100 mg of PMMx in 1 mL of THF was added into 6 mL of 0.1 M KOH methanol and stirred at 25 °C. At certain intervals, 1 mL of the mixed solution was taken out and neutralized with HCl. Afterwards, solvents were evaporated and the residue (polymer and KCl) was dissolved in chloroform and water. The chloroform solution was taken and solvent evaporation was done to obtain the degraded product of PMMx, which is analyzed with GPC.

3.6. Enzymatic degradation

The enzymatic degradation was measured by QCM-D from Q-sense AB (Sweden). PMMx film was prepared by spin-casting from polymer solution in THF (5.0 mg/mL) on AT-cut quartz crystal surface, and then dried at 60 °C for 24 h. The quartz crystal was placed in a fluid cell with the side of polymer film exposed to the solution. The lipase solution (1.0 mg/mL) was delivered to the surface at a flow rate of 150 $\mu\text{L}/\text{min}$ using ASW as reference. Details of the QCM-D measurements can be found elsewhere [33]. Briefly, the shift in frequency (Δf) of the crystal is

Table 1
Characterization data of PMMx.

Sample	Feed molar ratio MDO/MMA	Molar composition ^a Polyester/MMA	M_n (kDa) ^b	PDI ^b	T_g (°C) ^c
PMM0	0/100	0/100	22	2.6	113
PMM10	20/80	10/90	27	2.0	56
PMM31	50/50	31/69	26	2.0	17
PMM49	65/35	49/51	19	2.3	6

^a Determined by $^1\text{H-NMR}$.

^b Determined by GPC.

^c Determined by DSC.

related to the mass change of the thin layer on quartz crystal, whereas the dissipation factor is related to the viscoelastic properties of the additional layer. All experiments were performed at 25 °C and all the data used were from the third overtone ($n = 3$).

3.7. Release rate of antifoulant

The release rate of DCOIT was determined by measuring its concentration using high performance liquid chromatography (HPLC) under a static condition. The coatings (95 wt% of polymer with 5 wt% of DCOIT) were prepared on an epoxy resin panel ($25 \times 75 \text{ mm}^2$), and then the panel was immersed in artificial seawater (ASW) at 25 °C. After a given period of time, the panel was transferred from the holding tank to an individual measuring container with 200 mL of fresh ASW. After immersion for 24 h, 10 mL of ASW was taken from the individual measuring container and extraction was done using 10 mL of dichloromethane. After drying under nitrogen gas, the extract was dissolved in 1 mL of methanol and then subjected to HPLC analysis using a reversed-phase system (Agilent 1100) with an Eclipse Plus C18 column ($2.1 \times 150 \text{ mm}$, $5 \mu\text{m}$) connected to a photodiode array detector at 285 nm. The DCOIT peak was determined from the retention time and its concentration was calculated from the established standard curves. The release rate of DCOIT from the film was calculated from the DCOIT concentration in ASW using Eq. (1),

$$R = C \times 20 / (D \times A) \quad (1)$$

where R is the release rate of DCOIT ($\mu\text{g cm}^{-2} \text{d}^{-1}$), C is the DCOIT concentration in methanol ($\mu\text{g mL}^{-1}$) calculated from HPLC data, D is the immersion time in the individual measuring container (1 day) and A is the surface area of coated film (18.75 cm^2).

3.8. Marine field tests

The field tests were performed from April 2017 to July 2017 in Xiamen Bay, China ($118^{\circ}9' \text{ E}$, $24^{\circ}33' \text{ N}$). The samples (95 wt% of polymer with 5 wt% of DCOIT) were applied onto the glass fiber reinforced epoxy resin panels ($100 \times 100 \times 3 \text{ mm}^3$) and then lowered into seawater at a depth of $\sim 1 \text{ m}$. After a certain period of time, the panels were taken out of the sea and carefully washed with seawater and photographed, and then they were immediately placed into the seawater to continue the test. An epoxy panel as control and panels coated with PMMx were tested. The anti-biofouling activity was evaluated according to ASTM D6990-05 (2011) [34]. The range of the fouling rating (FR) was from 0 to 100, and the FR for a coating system free of adherent biofouling settlement shall be recorded as 100. The fouling rating for a paint film free of macrofouling settlement but partially or totally covered by microfouling growth (adherent slime) shall be recorded as 99, irrespective of the percent area covered by the “adherent slime”. Upon settlement of macrofouling forms, the total sum of percentage of area covered by macrofouling shall be deducted from 100. The fouling rating essentially reflects non-fouled area.

4. Result and discussion

The adhesion strength of biodegradable polymer to substrate is critical for its application in marine environment. We measured the adhesion of PMMx on an epoxy panel (Fig. 1). For PMM0, the adhesion strength value is about 3.1 MPa. As the content of main-chain esters increases, the adhesion gradually decreases. For PMM10, PMM31 and PMM49, the adhesion strength values are about 2.8, 1.9, 1.8 MPa. This is because the density of polar acrylate groups decreases as main-chain ester content increases, which reduces the polar interaction between the polymer and the substrate. Anyhow, all the coatings exhibit good adhesion with a value higher than 1.5 MPa which can meet the needs for application in marine environment.

In marine environment, there are various microorganisms such as

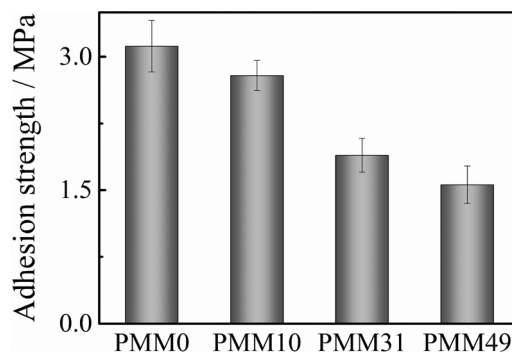


Fig. 1. Adhesion strength of the PMMx to epoxy panel.

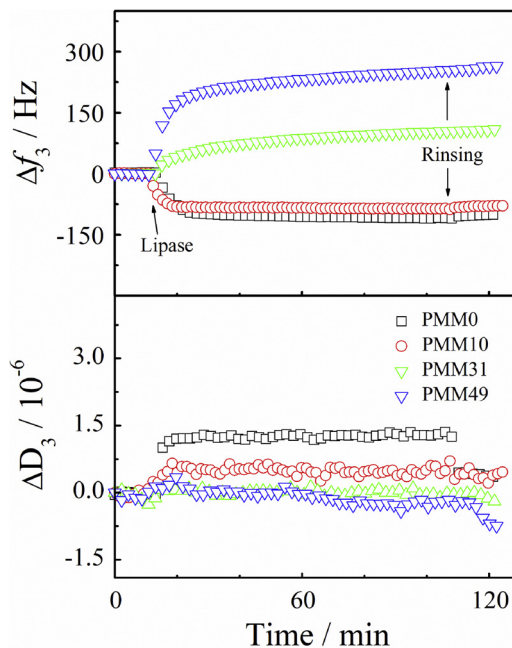


Fig. 2. Time dependence of the frequency shift (Δf) and dissipation shift (ΔD) regarding the enzymatic degradation of PMMx in ASW at 25 °C.

bacteria, actinomycetes and fungi, which can secrete enzymes and induce biodegradation. We examined the enzymatic degradation of PMMx by QCM-D. Fig. 2 shows the time dependence of the frequency shift (Δf) and dissipation shift (ΔD) regarding the enzymatic degradation of PMMx in ASW at 25 °C. For PMM0, Δf decreases after the lipase PS is introduced and gradually levels off. After rinsing with ASW, Δf still exhibits a notable decrease relative to the baseline. It is known that the increase in mass on the sensor surface causes the frequency to decrease. The decrease in Δf indicates the adsorption of lipase PS on PMM0 film. For PMM10, Δf decreases relative to the baseline but is smaller than that of PMM0 since the degradation is very slow and the adsorption of lipase PS causes the mass to increase. For PMM31 and PMM49 with higher main-chain ester contents, Δf exhibits notable increase relative to the baseline after rinsing, indicating mass loss of the films. This is because the copolymer chains decomposes into small fragments and the film surface disperse into the solution. Clearly, the degradation rate increases with main-chain ester content as reflected by the higher increase in Δf . On the other hand, ΔD increases with the thickness but decreases with the rigidity of a layer. Namely, a dense and rigid structure leads to a small dissipation of energy, whereas a looser and thicker structure results in a larger dissipation. For PMM0, the marked increase in ΔD relative to the baseline further indicates that lipase is adsorbed, forming a viscoelastic layer. For PMM10, the increase in ΔD is lower than PMM0, this is because degradation leads to a thin layer

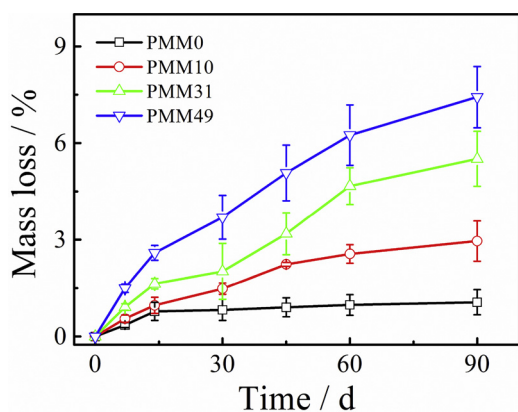


Fig. 3. Time dependence of mass loss of PMMx in ASW at 25 °C.

but lipase adsorption leads to a thick layer and the latter is more significant. PMM31 and PMM49 with the higher main-chain ester contents show a relatively less increase in ΔD , indicating a lower thickness resulted from fast degradation.

Fig. 3 shows the time dependence of mass loss of PMMx in ASW at 25 °C. For PMM0, it has some mass loss at the initial stage, which is probably caused by a quick release of residual solvent or some small molecules. Two weeks later, PMM0 does not show obvious variation of mass loss because it does not contain ester units in the main-chain and the C-C backbone is stable under this condition. For PMM10, PMM31 and PMM49, their weights have kept decreasing since they are immersed in ASW, this is because the cleavage of main-chain ester bonds decompose the polymeric surface into small fragments which subsequently disperse into ASW. Moreover, the mass loss increases with main-chain ester content. For PMM10, PMM31 and PMM49, the mass loss is 3.0%, 5.5% and 7.4% after 90 days, respectively. In other words, the hydrolytic degradability of PMMx is tunable by the density of main-chain esters, which well correlates with the enzymatic degradation results. Besides, the difference in T_g of polymers also influences the degradation because water diffusion is faster when the T_g is lower, facilitating the hydrolysis of ester bonds [35]. Nevertheless, the content of main-chain ester plays a more important role in degradation because it determines the number of spots available for main-chain cleavage, which contributes much more than that of T_g [16].

We also studied the degradation product of PMMx under an accelerated condition. As shown in Fig. 4(a), the molar mass of PMM0 does not change within the time range of the test, indicating that the C-C backbone is stable even in a strong alkaline solution. In comparison, PMM10, PMM31 and PMM49 exhibit a quick decrease in molar mass due to the cleavage of ester. The final degraded product are oligomers

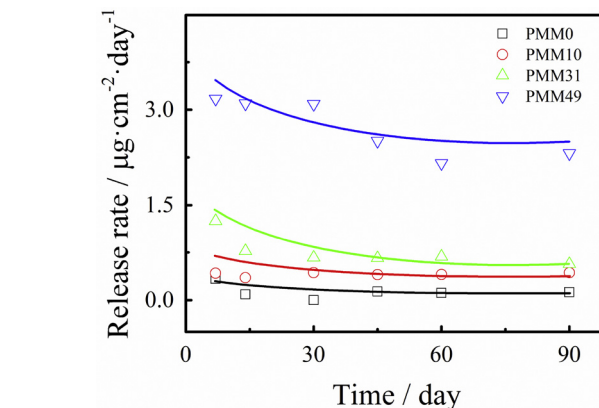
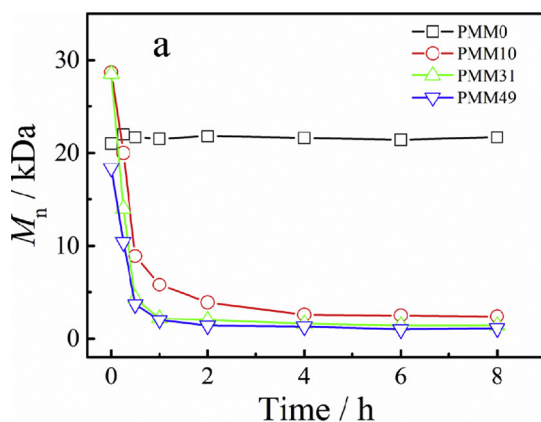


Fig. 5. Time dependence of the release rate of DCOIT from PMMx in ASW at 25 °C.

with average molar mass of 1.0–2.5 kDa, which decreases as main-chain ester content increases, showing that MMA and polyester moieties are not completely random in PMMx. This is because MMA and MDO monomers have different reactivity as shown in Fig. 4(b), where MMA reacts faster than MDO during the copolymerization. This leads to their slight uneven distribution in the polymer chains and some consecutive MMA repeating units exist. Thus, the final degradation products are PMMA oligomers. Although traditional polyacrylate can lead to marine plastic pollution, it was reported that PMMA does not have toxicity to marine organisms [36,37], the degraded PMMA oligomers which are easier to metabolize should not pose threat either. Thorough toxicity test will be performed in our future work.

Biodegradable polymer itself would have a good antifouling performance if its degradation rate is enough high. However, high degradation rate leads to an inevitably short service life. Hence, it can be a good strategy to enhance the antifouling duration of biodegradable polymer by adding eco-friendly antifoulant. Fig. 5 shows the time dependence of release rate of DCOIT from PMMx in ASW at 25 °C. Each PMMx film has a relatively high release rate in the early stage, but the rate gradually decreases and levels off. For PMM0, the release rate is always low during the test period because PMM0 is an inert matrix and the release of DCOIT is only a slow diffusion process. For PMM10, PMM31 and PMM49, their release rates are higher than PMM0 because polymer degradation leads to a renewed surface and DCOIT could be readily released into ASW. Moreover, the release rate increases as the main-chain ester content increases. After immersion in ASW for 1 month, the release rates are stable at about $0.6 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$, $0.8 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$ and $2.5 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{day}^{-1}$, respectively. Thus, we can regulate DCOIT release by varying the density of main-chain esters.

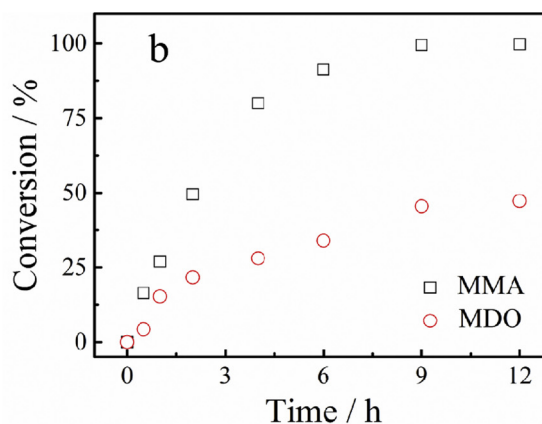


Fig. 4. Time dependence of molar mass of PMMx in 0.1 M KOH methanol solution at 25 °C (a). Time dependence of monomer conversion during MMA-MDO copolymerization at a molar feed of 50/50 (b).

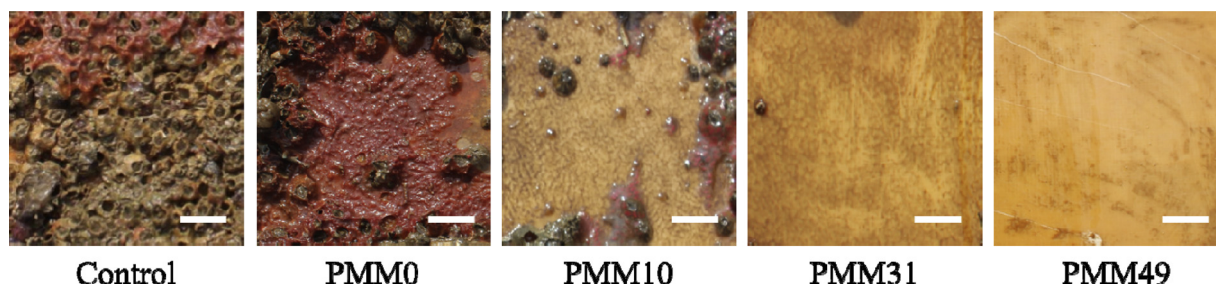


Fig. 6. Images of PMMx coatings with 5 wt% of DCOIT after immersion in seawater for 3 months (April–July 2017, Xiamen Bay, China). Scale bar: 2 cm.

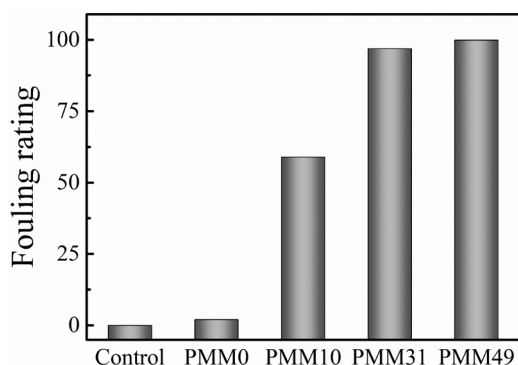


Fig. 7. Fouling rating of PMMx coatings with 5 wt% of DCOIT after immersion in the seawater for 3 months (April–July 2017, Xiamen Bay, China).

Clearly, degradation of the copolymers leads to renewal of the surfaces and the constant release of antifoulants, which is beneficial for prolonged anti-biofouling ability.

The anti-biofouling performance of the system consisting of PMMx and DCOIT was evaluated by marine field test. The typical images of tested panels after immersed in seawater for three months are shown in Fig. 6. The corresponding fouling rating is shown in Fig. 7. The control epoxy panel is seriously fouled by marine organisms and its fouling rating is 0, indicating a high fouling pressure during the test. PMM0 also exhibits a poor anti-biofouling efficiency because its surface is not renewable and the released DCOIT is not enough to inhibit the fouling. For PMM10, some marine organisms can be found on the surface after immersion for three months but its anti-biofouling performance is better than PMM0. This is because the degradation of copolymer generates a renewed surface which polishes the attached living organisms or inorganics away, and the higher release rate of DCOIT than PMM0 has a higher efficiency in inhibiting the biofouling. PMM31 and PMM49 exhibits better anti-biofouling ability with fouling rating higher than 95 because their degradations are faster and the antifoulants are released at higher rates. In short, a system with reasonable degradation rate and controlled release of antifoulants would have good anti-biofouling ability.

5. Conclusion

We have prepared biodegradable poly(ester)-poly(methyl methacrylate) copolymer by a one-step radical ring-opening copolymerization. Such copolymer coating has good adhesion to substrate and tunable enzymatic and hydrolytic degradation. The copolymer has a controlled release rate of DCOIT by varying the composition. The system consisting of biodegradable polymer and DCOIT exhibits good antifouling performance and is expected to find applications in marine anti-biofouling.

Acknowledgements

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